Scalability & Future Plan

Date	01-11-2026
Team ID	EN25CS3011097
Project Name	Global Olympic Sports Performance Analysis

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static CSV data covering 18 Games (1992-2026) with no live feed	Dashboard does not update automatically when new results arrive	High
2	No geographic map view; analysis limited to bar charts	Regional patterns are harder to see at a glance	Medium
3	Four record types in one table require filters in every view	Risk of incorrect totals if a filter is missed	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Packaged workbook used by the team and evaluators	Publish on Tableau Public / Tableau Server so many viewers can use it at once
2	Data Storage	8,988 rows in a CSV and a Snowflake table	Store all Games in Snowflake with separate tables per record type
3	Performance	Extract-based dashboard renders quickly	Use scheduled extract refresh and aggregate tables; scale the Snowflake warehouse size when needed
4	Security	No personal sensitive data; access limited to the team	Role-based access in Snowflake and Tableau, with secured shared links

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Geographic map of medals and Games-year trend lines	Next 1 month	Shows regional and historical patterns
Phase 3	Live Snowflake connection with scheduled refresh; efficiency (medals per athlete) and GDP-tier analysis	Next 2-3 months	Always current data and deeper insight
Phase 4	Public dashboard with mobile layout; forecast of medal counts	Next 4-6 months	Wider audience and predictive insight